

Seismic Loading Framework Tutorial

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1. Consider two buildings, A and B , both with design lives of 50 years. Building A is a generic office building and should be designed for a level of ground-motion that has a 10% chance of being exceeded over its life. Building B is a critical facility for which the consequence of failure is very high. The probability of exceedance for the design level ground-motion for Building B should only be 1%.

(a) Compute the ratio of the return periods for buildings B and A .

Solution: Assuming exceedances are represented by a Poisson distribution, the key expression is:

$$P(n \geq 1) = 1 - e^{-\lambda(IM > im) \times T_{DL}}$$

Solving for $\lambda(IM > im)$ gives:

$$\lambda(IM > im) = -\frac{\ln[1 - P(n \geq 1)]}{T_{DL}}$$

The return period is defined as $RP = 1/\lambda(IM > im)$, so for building A we have:

$$\lambda_A(IM > im) = -\frac{\ln(1 - 0.1)}{50} = 2.107 \times 10^{-3} \text{ years}^{-1}$$

$$RP_A = \frac{1}{2.107 \times 10^{-3}} = 474.6 \text{ years}$$

For building B we have:

$$\lambda_B(IM > im) = 2.01 \times 10^{-4} \text{ years}^{-1}$$

and

$$RP_B = \frac{1}{2.01 \times 10^{-4}} = 4975 \text{ years}$$

The ratio is therefore:

$$\frac{RP_B}{RP_A} = \frac{4975}{474.6} = 10.48$$

- (b) If the average annual rates of exceedance of ground-motion levels im is defined by:

$$\lambda(IM > im) = 0.05 \exp(-2im)$$

compute the corresponding ratio of design ground-motion levels for the two buildings.

Solution: The hazard curve is of the form: $\lambda(IM > im) = \lambda_0 \exp(\alpha \times im)$, and we can invert this to obtain im as a function of $\lambda(IM > im)$.

$$im = \frac{1}{\alpha} \ln \left[\frac{\lambda(IM > im)}{\lambda_0} \right]$$

For our particular case, we have $\lambda_0 = 0.05$ and $\alpha = -2$. Therefore, for building A we

have:

$$im_A = -\frac{1}{2} \ln \left(\frac{2.107 \times 10^{-3}}{0.05} \right) = 1.58$$

For building B we have:

$$im_B = -\frac{1}{2} \ln \left(\frac{2.01 \times 10^{-4}}{0.05} \right) = 2.76$$

And the corresponding ratio is:

$$\frac{im_B}{im_A} = \frac{2.76}{1.58} = 1.75$$

2. Imagine that the vulnerability of a building can be expressed in terms of the way that the probability of failure, P_f , varies with resistance, r . Further imagine that a particular relationship between these quantities is:

$$P_f(r) = \exp(-r/2)$$

and assume that the cost of failure is 1000 monetary units.

A given building system, $S1$, can provide a level of resistance according to a cost function:

$$C_{C,S1}(r) = 100 + 5 \exp(2r)$$

- (a) Determine the optimal level of resistance for this building and the associated total cost.

Solution: The cost of failure can be expressed as:

$$C_F(r) = \alpha_F \exp(\beta_F r) = 1000 \exp(-r/2)$$

The cost of construction is:

$$C_{C,S1}(r) = C_{C0} + \alpha_C \exp(\beta_C r) = 100 + 5 \exp(2r)$$

The total cost can therefore be written generically as:

$$C_{T,S1} = C_F(r) + C_{C,S1}(r)$$

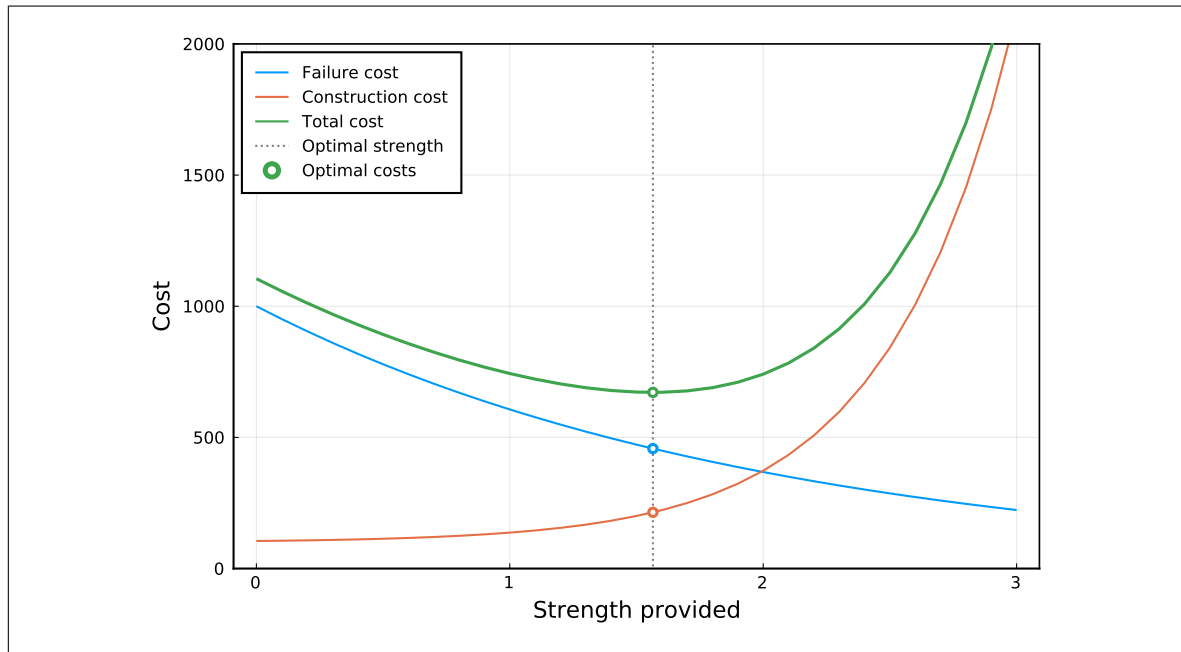
The optimal level of resistance r^* can be obtained by differentiating the above function, setting the derivative to zero, and solving for r .

$$\frac{dC_{T,S1}}{dr} = \alpha_F \beta_F \exp(\beta_F r) + \alpha_C \beta_C \exp(\beta_C r)$$

when

$$\frac{dC_{T,S1}}{dr} = 0 \quad \rightarrow \quad r^* = \frac{1}{(\beta_F - \beta_C)} \ln \left(-\frac{\alpha_C \beta_C}{\alpha_F \beta_F} \right)$$

For $\alpha_F = 1000$, $\beta_F = -0.5$, $\alpha_C = 5$ and $\beta_C = 2$, we have $r^* = 1.565$. This solution can be visualised in the following figure.

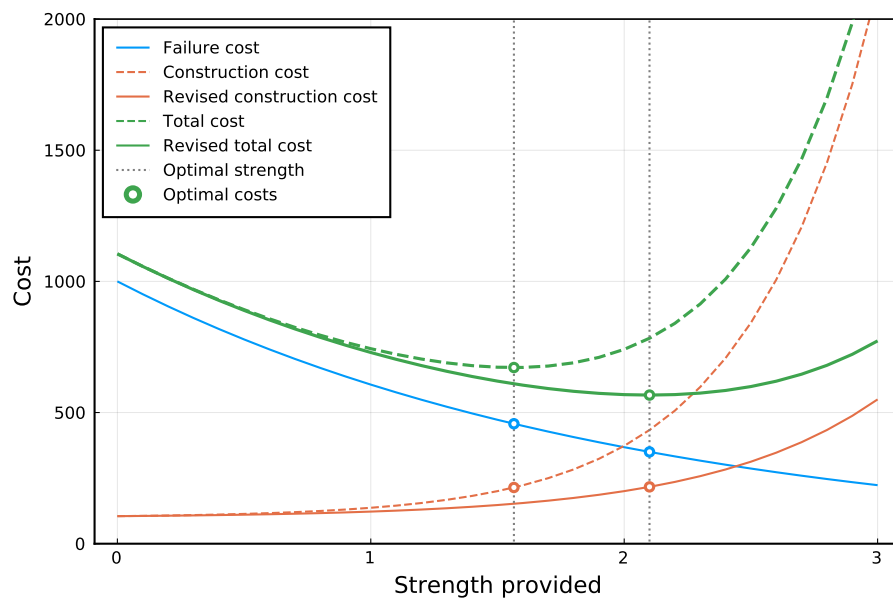


(b) Another building system, S_2 , can provide the same level of resistance for cheaper:

$$C_{C,S2}(r) = 100 + 5 \exp(1.5r)$$

What is the reduction in total cost associated with using system S_2 over S_1 ?

Solution: The process for obtaining the optimal solution for this system is the same as in the above part. In this case, the optimal strength comes out at $r^* = 2.1$, as shown in the figure below:



The total cost for system S_1 was:

$$C_{T,S1} = C_F(r^* = 1.565) + C_{C,S1}(r^* = 1.565) = 672$$

The total cost for system S_2 is:

$$C_{T,S2} = C_F(r^* = 2.10) + C_{C,S2}(r^* = 2.10) = 567$$

The reduction in the total cost is therefore $\Delta C_T = 105$, or

$$\frac{C_{T,S2} - C_{T,S1}}{C_{T,S1}} = -15.6\%$$

- (c) What is the change in the probability of failure if this alternative system $S2$ is adopted over system $S1$?

Solution: The change in the probability of failure is found from first computing the probabilities of failure associated with each optimal solution.

$$P_f(r) = \frac{C_F(r)}{\alpha_F}$$

Therefore:

$$P_{f,S1}(r^* = 1.565) = 45.7\%$$

$$P_{f,S2}(r^* = 2.10) = 35.0\%$$

The use of system $S2$ over $S1$ thus leads to a 23.5% reduction in the probability of failure.

$$\frac{P_{f,S2} - P_{f,S1}}{P_{f,S1}} = -23.5\%$$